Week 9: 26/10/2026 – 30/10/2026

Subject: Mathematics Course 3 **Unit Title: Exploring Transformations and Linear Relationships** **Grade : 8**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: Slope and similar triangles-Part 1	I can understand how similar right triangles can be used to calculate and compare slope on a straight line.	1.1 Flexible display of information (size, color, font, layout). 2.3 Support decoding of text and notation using tools, audio, and multiple representations. 8.1 Make goals clear, visible, and supported by short-term objectives.	MAGMA MATH: Students will check the posted assignments /lessons on Magma Girls 'campus: Students will check their materials on Files section on Teams.
Lesson 2: Slope and similar triangles-Part 2	I can justify why slope remains constant along a straight line using similar triangles and real-life UAE applications.		
Lesson 3: Formative	For Girls 'campus FORMATIVE (NOT SUMMATIVE) on Thursday.29 October in the following two main topics: 1.Transformations 2.Slope and straight lines (lessons included from the textbook: 2.1-2.2-2.3 pages:53-70 3.1-3.2 pages:115-126) Check the toddle announcement and teams for the materials		
Lesson 4: Solve linear equations	I can solve linear equations.		

Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Make sure your devices are fully charged. Bring your pencil case/calculator with you. Complete your extra practice on Magma Math App. **As per school policy, students can have a formative and a summative in the same day.** **Grade 8B math summative will be in week 10:** **Girls'campus:Thursday.5.11** **Boys'campus: Tuesday, November 3rd ,2026** **Study guide and review sheet for the summative will be shared on toddle/teams within Monday: 26 October 2026.**

Technology Integration دمج التكنولوجيا	Resources shared on Teams
Cross-curricular links الروابط بين المواد الدراسية	Science Connection: Students explore how slope relates to the steepness of inclined planes and how it affects the force needed to move objects up a ramp. IT Connection: Students use GeoGebra or Desmos to construct similar triangles on a straight line and investigate how the rise/run ratio remains constant.
External Assessment Integration دمج التقييمات الخارجية	All content is covered in MAP
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	🧠 **Wellbeing** • Confidence in solving real-world measurement problems. • Resilience when learning from mistakes. • Collaboration through shared investigations and discussion. • Self-awareness by checking the reasonableness of solutions. 🌍 **IB ATL Skills** **Thinking:** solving problems using scale factors and proportional reasoning. **Communication:** explaining mathematical methods and solutions clearly.

	Research: interpreting scale drawings and real-world models. **Self-Management:** organising calculations accurately and efficiently. **Social:** discussing strategies and comparing solutions with others
Standards المعايير	8.F.A.1: Understand that a function is a rule that assigns each input exactly one output. 8.F.B.4: Construct a function to model a linear relationship between two quantities and interpret the rate of change and initial value MP3: Construct viable arguments and critique the reasoning of others. MP7: Look for and make use of structure.
Inquiry/ TOK Question أسئلة البحث والاستقصاء TOK	How does the accuracy of a scale factor affect the final size and reliability of a real-world design? How can slope help us understand and compare rates of change in real-life situations? How can a graph help us identify and understand the unit rate in a proportional relationship? How can unit rates help us compare proportional relationships represented in different ways?
Success Criteria معايير النجاح **(Measurable)**	<u>Lesson 1:</u> 🟦 All students must be able to identify the rise and run of right triangles formed along a straight line. 🔵 Most students should be able to calculate the slope of a line using the rise/run ratio of similar triangles. 🔺 Some students could be able to explain why different-sized similar triangles on the same line have equal slopes <u>Lesson 2:</u> 🟦 All students must be able to identify similar triangles on a straight line and calculate their slopes. 🔵 Most students should be able to demonstrate that the slope remains constant by comparing the rise/run ratios of different similar triangles. 🔺 Some students could be able to justify why slope remains constant using triangle similarity and apply their reasoning to UAE ramps, roads, or rail routes.

	Lesson 3: Formative will be in the following lessons from the textbook: 2.1-2.2-2.3 pages:53-70 3.1-3.2 pages:115-126 Lesson 4: ■ All students must be able to solve one step linear equations. ● Most students should be able to solve multi-step linear equations. ▲ Some students could be able to translate a reallife scenario into a linear equation.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Belonging الإنتماء Respect الاحترام Heritage التراث
	Students connect slope and proportional relationships to **UAE roads, transport routes, and infrastructure**, strengthening their understanding of how mathematics supports their community. Students demonstrate respect by **collaborating, sharing mathematical strategies, and appreciating different approaches** when solving UAE real-world problems. Students explore proportional relationships through **UAE heritage contexts**, such as traditional architecture, falaj irrigation systems, and historical trade routes.